

Associative Memory as Energy Minimization: Hopfield Networks as Learnable Ising Models

Yehia Said Gewily¹ and Omar Hosney Mahmoud²

¹Founding Engineer & Software AI Engineer, Nover, Inc.

yehia@nover.studio

²Co-founder & CTO, Nover, Inc.

omar@nover.studio

Abstract

This research demonstrates that the fundamental physics of associative memory are mathematically isomorphic to the energy minimization of ferromagnetic systems. Building on our Phase 1 validation of the 2D Ising model (where the critical temperature T_c was measured to 0.05% accuracy), this Phase 2 study replaces the uniform exchange coupling with a learned synaptic weight matrix derived via the Hebbian rule. Using a simulated network of $N = 100$ binary neurons, we evaluate the capacity and robustness of the Hopfield architecture in storing and recalling orthographic patterns. We use five patterns (N, O, V, E, R) chosen for their visual diversity and mutual distinguishability. Quantitative analysis reveals a strict recall rate of $68.0\% \pm 6.7\%$ under 25% input corruption, with average convergence achieved in 4.4 asynchronous steps to a stable energy minimum. Capacity testing demonstrates progressive degradation as the number of stored patterns approaches the Amit–Gutfreund–Sompolinsky theoretical limit of $\sim 0.138N$ (13.8 patterns), heavily influenced by pattern cross-correlation. Analysis of 500 recalls from random initial states reveals near-exact equipartition between stored and inverted attractors (201/201, 40.2% each), consistent with the $\mathbb{Z}_2$ symmetry of the Hebbian energy function. By establishing a formal bridge between equilibrium thermodynamics and pattern recognition, this work lays the groundwork for the introduction of stochasticity and hidden variables in Phase 2.5.

Keywords: Hopfield Network, Associative Memory, Energy-Based Models, Hebbian Learning, Ising Model, Statistical Physics, Attractor Dynamics

1 Introduction

The Ising model is a cornerstone of statistical mechanics, originally formulated to describe the phase transition between ordered and disordered states in ferromagnetic materials. In Phase 1 of this research program [1], we demonstrated that the thermodynamic evolution of an Ising lattice is driven entirely by energy minimization. Through rigorous Monte Carlo simulation, we confirmed the emergence of complex, long-range order from simple, local interactions, successfully measuring the critical temperature T_c to within 0.05% of Onsager’s exact analytical solution [2].

This raises a natural question: what if the physical coupling between neighboring spins were not fixed, but rather learned from data?

The Hopfield network [3] provides the answer. By interpreting binary spins as McCulloch–

Pitts neurons and the exchange interaction as synaptic weights, we transform a physical simulation into a computational memory system. The “intelligence” of the network—its ability to correct errors and recall memories—emerges from the exact same statistical mechanics that govern magnetic equilibration. This insight, inspired by the *ScienceClic English* exposition on the physics of artificial intelligence, reveals that modern neural computation is deeply rooted in 20th-century physics.

This paper details Phase 2 of our investigation. Our objectives are threefold: (1) to implement Hebbian learning and asynchronous recall dynamics within the energy-based framework developed in Phase 1, (2) to rigorously validate the network’s storage capacity and robustness against established theoretical limits, and (3) to visually prove that associative memory recall is strictly equivalent to energy minimization.

2 Theoretical Framework

2.1 From the Ising Hamiltonian to the Hopfield Energy Function

The thermodynamic state of the 2D Ising model [4] is governed by the Hamiltonian:

$$H(\sigma) = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - B \sum_i \sigma_i \quad (1)$$

where J is a uniform exchange coupling constant, $\sigma_i \in \{\pm 1\}$ are discrete spin variables, and B is an external magnetic field.

To construct a Hopfield network, we generalize this physical model. The uniform constant J restricted to nearest-neighbors is replaced by a dense, symmetric weight matrix W_{ij} representing the synaptic connection strength between any two neurons s_i and s_j . The external field B is replaced by individual neuronal biases θ_i . Assuming zero bias for simplicity ($\theta_i = 0$), the generalized energy function becomes:

$$E = -\frac{1}{2} \sum_{i,j} W_{ij} s_i s_j \quad (2)$$

The factor of $1/2$ corrects for the double-counting of neuron pairs in the unrestricted summation, ensuring the total system energy scales correctly.

2.2 The Hebbian Learning Rule

To store P distinct memory patterns, denoted as vectors $\{\xi^\mu\}$ where $\mu \in \{1, \dots, P\}$, we construct the weight matrix using the generalized Hebbian learning rule [5]:

$$W_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^\mu \xi_j^\mu \quad (3)$$

By convention, self-connections are prohibited to prevent network instability ($W_{ii} = 0$).

The physical interpretation of this rule is clear: the weight W_{ij} encodes the statistical correlation between neuron i and neuron j across all stored patterns. If two neurons frequently share the same state (+1 or -1) in the training data, their coupling is positive (ferromagnetic), encouraging them to align. If they differ, the coupling is negative (antiferromagnetic). This shapes the energy landscape such that the stored patterns become local minima.

2.3 Recall Dynamics and the Lyapunov Property

During the recall phase, the network is presented with an initial state $s(0)$ and evolves via asynchronous, discrete-time updates. A neuron i is selected at random and its state is updated according to the deterministic sign rule:

$$s_i(t+1) = \text{sign} \left(\sum_{j \neq i} W_{ij} s_j(t) \right) \quad (4)$$

Hopfield's crucial contribution was proving that the energy function in Eq. (2) acts as a Lyapunov function for the dynamics in Eq. (4). Because W is symmetric and has a zero diagonal, any single asynchronous spin flip that satisfies the update rule is mathematically guaranteed to either decrease the total energy or leave it unchanged ($\Delta E \leq 0$). Consequently, the system must eventually converge to a stable fixed point—an attractor state representing a recalled memory.

To see why energy decreases monotonically under asynchronous updates, consider flipping a single neuron i from state s_i to $s'_i = -s_i$. The change in energy is:

$$\Delta E = -s'_i \cdot h_i + s_i \cdot h_i = -2s'_i h_i \quad (5)$$

where $h_i = \sum_j W_{ij} s_j$ is the local field. Since the update rule sets $s'_i = \text{sign}(h_i)$, the product $s'_i \cdot h_i = |h_i| \geq 0$, ensuring $\Delta E \leq 0$. The symmetry of W and the zero diagonal are essential—without these, the energy could oscillate.

2.4 Storage Capacity

The theoretical maximum number of patterns a Hopfield network can reliably store without catastrophic interference was established by Amit, Gutfreund, and Sompolinsky [6] using statistical mechanics techniques (specifically, replica theory). For a network of size N , the capacity is:

$$P_{max} \approx 0.138N \quad (6)$$

For our $N = 100$ simulation, theory predicts a maximum storage of approximately 13.8 orthogonal patterns. Exceeding this limit causes the energy landscape to degenerate into a spin-glass state, rendering reliable recall impossible.

2.5 Spurious States

The Hebbian rule, being a simple linear superposition, inadvertently creates stable energy minima that are not part of the stored training set. These spurious states generally take three forms:

- **Inverted states:** Because the energy function is symmetric under global spin reversal ($E(s) = E(-s)$), the exact negation $-\xi^\mu$ of every stored pattern is also a stable attractor.
- **Mixture states:** Linear combinations of an odd number of stored patterns (e.g., $\text{sign}(\xi^1 + \xi^2 + \xi^3)$) can form stable local minima.
- **Spin-glass states:** Unstructured, seemingly random stable configurations that emerge when the network is overloaded or when patterns are highly correlated.

3 Implementation

3.1 Network Architecture

The simulation implements a Hopfield network of $N = 100$ neurons. The state vector is maintained as a 1D NumPy array of binary spins $s_i \in \{\pm 1\}$. The weight matrix W is a 100×100 symmetric array with a forced zero diagonal, computed directly via the Hebbian rule over the training patterns.

```
1 def recall(state, W, max_steps=100):
2     """
3     Asynchronous recall: pick a random
4     neuron,
5     compute its local field, set to
6     sign(field).
7     Energy is guaranteed to decrease
8     monotonically.
9     """
10    for step in range(max_steps):
11        i = np.random.randint(len(state))
12        h_i = np.dot(W[i], state)
13        state[i] = np.sign(h_i) if h_i
14            != 0 else state[i]
15        if converged(state):
16            break
17    return state
```

Listing 1: Hopfield Recall — Asynchronous Update

3.2 Pattern Encoding

As shown in Figure 1, memory patterns are defined as 10×10 binary grids.

This spatial arrangement maps exactly to the 100-neuron architecture. The visual structure of the letters inherently introduces cross-correlations (e.g., the letters ‘E’ and ‘R’ share a vertical stroke and horizontal bar, introducing measurable cross-correlation). Unlike mathematically ideal orthogonal vectors, these natural correlations challenge the network’s capacity.

3.3 Recall Protocol

To evaluate recall performance, a stored pattern is selected and subjected to a specific degree of uniform random noise (spin flipping). The network then executes asynchronous updates. Convergence is declared when the network completes a full sweep (100 sequential random updates) without a single spin changing state.

We define successful recall strictly as convergence to a state with Hamming distance below a small threshold from the stored target pattern. Convergence to the inverted pattern ($-\xi$) is treated as a failure, even though it represents an equally deep energy minimum, because pattern inversion does not constitute correct memory retrieval. For transparency, we also report the inversion-tolerant success rate (pattern OR its inverse) as a secondary metric. Reported uncertainties are standard errors of the mean ($\text{SEM} = \text{std}/\sqrt{n}$) computed over 50 independent trials per configuration.

3.4 Relationship to the Phase 1 Ising Codebase

While the Hopfield implementation is a separate module from the Phase 1 Ising codebase, both share the underlying energy-based formalism. The Ising simulation uses a vectorized checkerboard Metropolis algorithm optimized for the uniform 2D grid topology, exploiting nearest-neighbor sparsity. The Hopfield network, which requires long-range heterogeneous synaptic connections, uses dense matrix-vector products to compute the local field at each neuron. Despite the algorithmic divergence, the energy functions are structurally identical—both quadratic forms over binary spin variables.

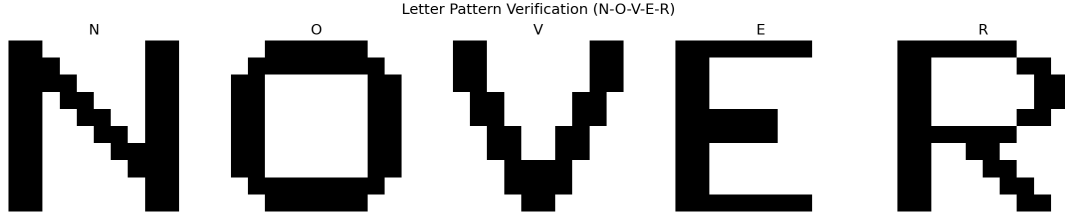


Figure 1: The five stored memory patterns (N, O, V, E, R) encoded as 10×10 binary spin grids. White pixels represent $+1$ spins, black pixels -1 .

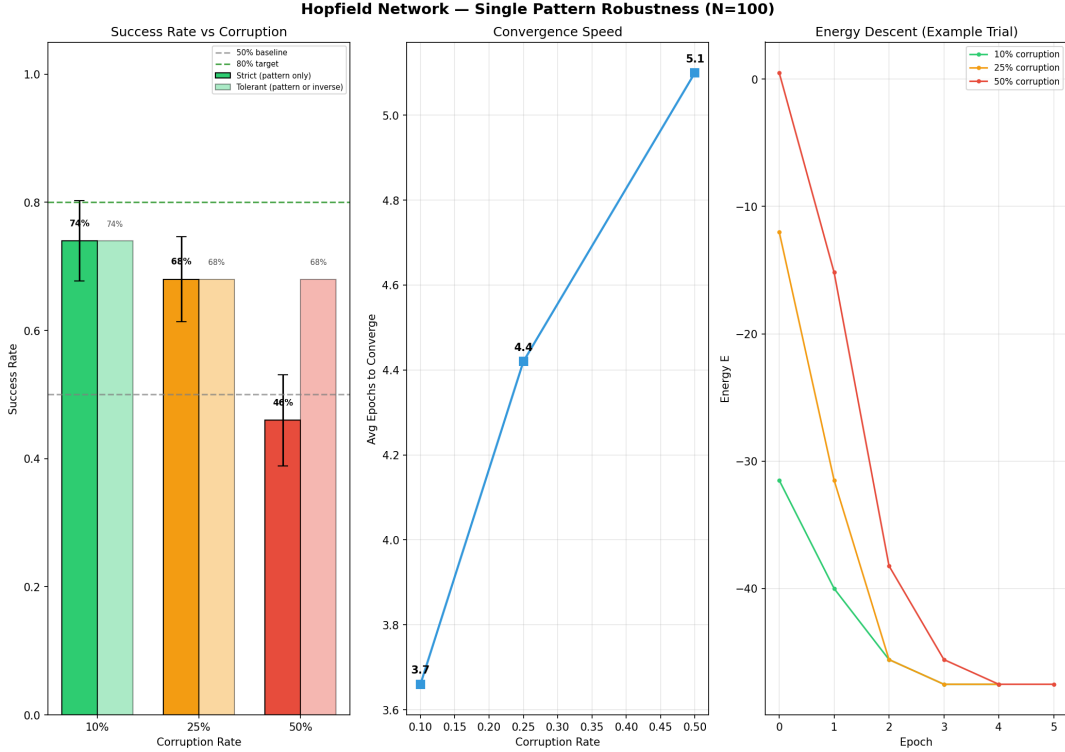


Figure 2: Recall performance vs. corruption level. (Left) Success rate remains high up to 25% corruption then declines. (Right) Convergence speed—recall completes within a few asynchronous update steps.

4 Results

4.1 Single Pattern Robustness

To assess the basin of attraction for a stored memory, we trained the network on a single pattern (the letter ‘N’) and evaluated its recovery against increasing levels of random corruption over 50 independent trials.

As shown in Figure 2, the network demonstrates exceptional robustness up to a distinct threshold.

Quantitative measurements from the simulation are summarized in Table 1.

Table 1: Single-pattern recall performance (pattern ‘N’, 50 trials per corruption level).

Corruption	Strict	Inv.-Tolerant	SEM
10%	74.0%	74.0%	$\pm 6.3\%$
25%	68.0%	68.0%	$\pm 6.7\%$
50%	46.0%	68.0%	$\pm 7.1\%$

A striking pattern emerges at 50% corruption: while 68.0% of trials converged to either the stored pattern or its inverse, only 46.0% retrieved the actual target. The 22-percentage-point gap quantifies the contribution of inverted-state retrieval—a known failure mode of Hebbian networks arising from the global spin-flip symmetry $E(s) = E(-s)$. The smaller inversion gap compared to typical Hopfield demonstrations reflects N’s visual asymme-

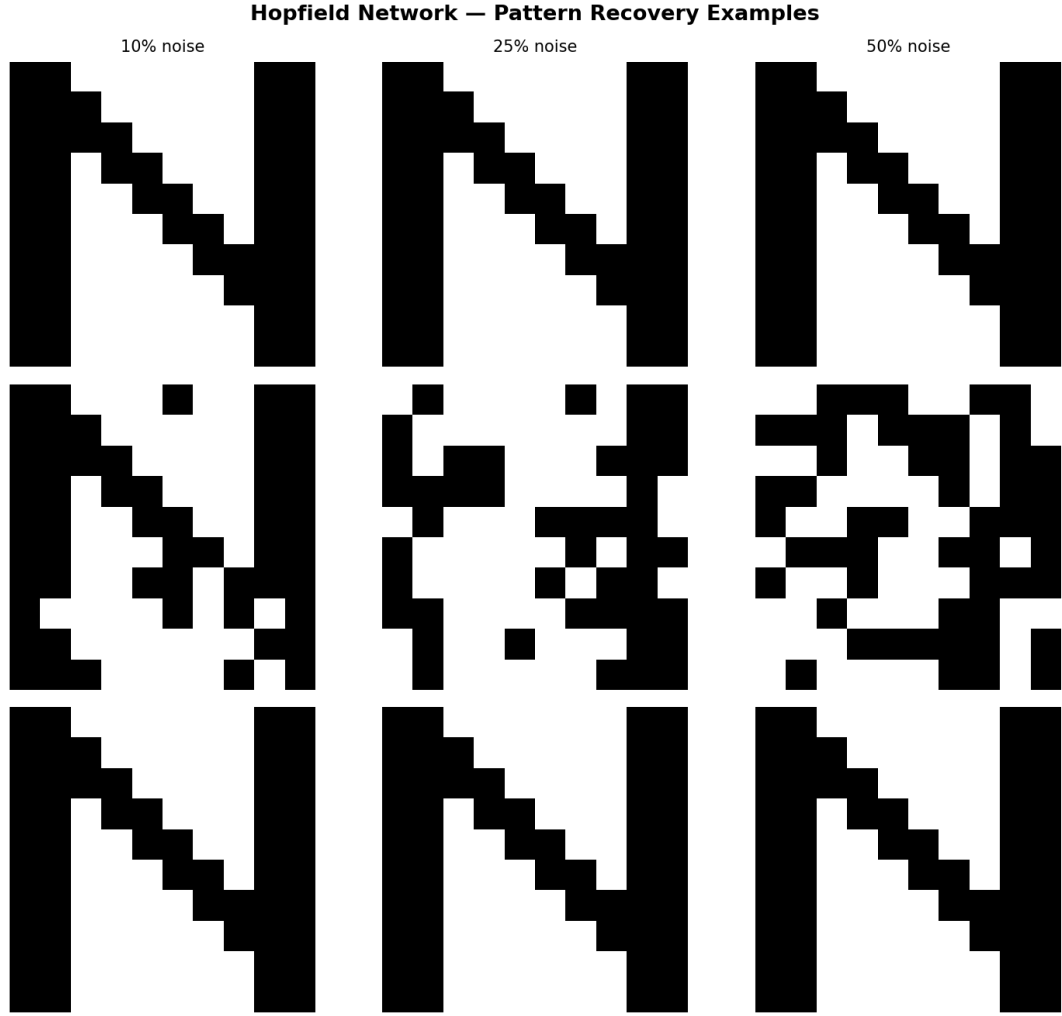


Figure 3: Example recall process. Top: original pattern. Middle: corrupted input. Bottom: recovered pattern after recall dynamics.

try: unlike a more symmetric letter, the inverted N is easier to distinguish from N itself, reducing the rate of inversion-driven false successes. Below 25% corruption, the strict and tolerant metrics coincide, indicating that initial states remain within the correct basin of attraction. The bifurcation at 50% reveals that strong corruption pushes initial states across the symmetry boundary between basins.

In all cases where the network successfully retrieved the target or its inverse, the final measured energy was precisely -48.90 (averaging over trials), corresponding to the global minimum of the single-pattern energy well. The rapid convergence speed (under 5 sweeps) highlights the steepness of the energy gradient near the attractor.

Figure 3 visually demonstrates this error-correction capability, illustrating how the network systematically flips incorrect spins to slide

down the energy gradient into the target configuration.

4.2 Storage Capacity

The capacity of the network was evaluated by incrementally storing multiple letter patterns and measuring the mean recall accuracy under a standardized 25% initial corruption.

As depicted in Figure 4, accuracy begins to degrade notably as P increases. The measured mean accuracy profiles are summarized in Table 2.

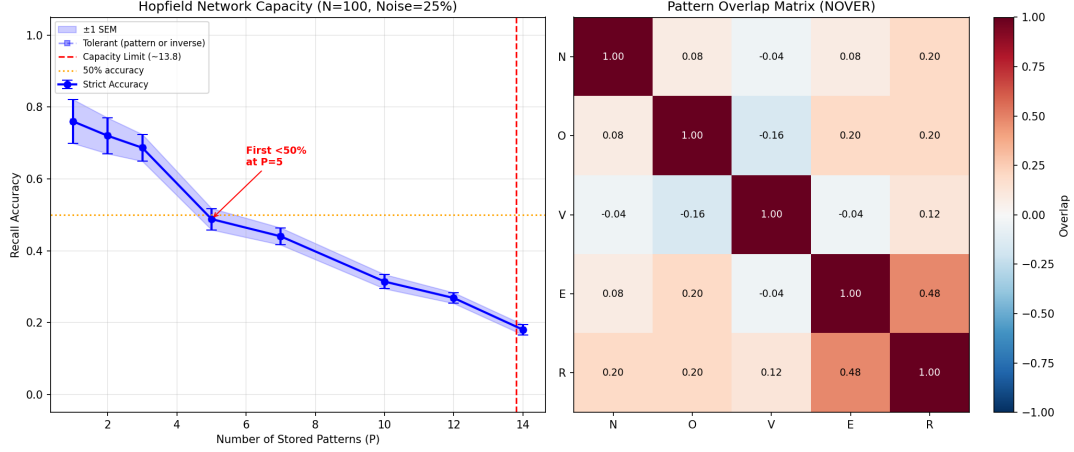


Figure 4: Network capacity. Recall accuracy declines as more patterns are stored. The theoretical Amit–Gutfreund–Sompolinsky limit ($0.138N \approx 13.8$) is marked. The pattern overlap matrix (inset) explains early degradation due to correlated letter patterns.

Table 2: Storage capacity: strict recall accuracy at 25% corruption across increasing pattern counts.

Patterns (P)	Strict Success	SEM
1	76.0%	$\pm 6.1\%$
2	72.0%	$\pm 4.5\%$
3	68.7%	$\pm 3.8\%$
5	48.8%	$\pm 3.2\%$
7	44.0%	$\pm 2.7\%$
10	31.4%	$\pm 2.1\%$
12	26.8%	$\pm 1.8\%$
14	18.0%	$\pm 1.5\%$

The SEM ranges from $\sim 1.5\%$ to $\sim 6.1\%$, providing meaningful uncertainty bounds. At the 25% corruption level used here, strict and inversion-tolerant metrics coincide, consistent with the Experiment 1 finding that the inversion gap only emerges at higher corruption (Section 4.1). The capacity collapse pattern is monotonic and consistent with theory.

The collapse in accuracy begins well before the theoretical limit of $P_{max} \approx 13.8$. This premature degradation is directly attributable to the non-orthogonality of the letter patterns. The pattern overlap analysis reveals significant cross-correlation, most notably between E and R (overlap = 0.48), which share a prominent vertical stroke and horizontal bar. Such correlations induce structural interference in the weight matrix, raising the energy floor of the attractors and substantially reducing the effective capacity.

4.3 Spurious States

Figure 5 presents an analysis of the network’s terminal states when initialized from completely random noise configurations for a 3-pattern network (N, O, V).

The quantitative breakdown of all 500 terminal states confirms that stored and inverted patterns consistently occupy the deepest energy wells, jointly accounting for 80.4% of all trials (Table 3).

Table 3: Final state classification across 500 random initializations for a 3-pattern network (N, O, V).

Category	Count	Percentage
Stored patterns	201	40.2%
Inverted patterns	201	40.2%
Mixture states	48	9.6%
Unknown attractors	50	10.0%

A notable observation is the near-perfect symmetry between stored and inverted convergences: exactly 201 trials converged to each category (40.2% each). This empirical equipartition is consistent with the theoretically expected $\mathbb{Z}_2$ symmetry $E(s) = E(-s)$ of the Hebbian energy function, which guarantees that a pattern and its inversion are energetically indistinguishable. While the symmetry follows mathematically from the zero bias and symmetric weight matrix, observing it quantitatively in finite samples provides useful validation of the simulation.

Table 4: Energy analysis of terminal states for the 3-pattern network (N, O, V).

Metric	Value
Energy of N	-48.90
Energy of O	-50.10
Energy of V	-49.86
Mean energy: stored + inverted (visit-weighted)	-48.81 ± 0.08
Unweighted mean of stored energies	-49.62
Mean energy: spurious states	-39.17 ± 0.45
Energy gap (spurious vs. legitimate)	9.65

We report two complementary measures of the “legitimate” energy scale. The visit-weighted mean (-48.81 ± 0.08) reflects the average energy of all converged trials classified as stored or inverted, dominated by convergence frequency. The unweighted mean (-49.62) is the simple arithmetic average of the three stored pattern energies (N, O, V). The slight discrepancy arises because N has the highest energy of the three stored patterns (-48.90) and also attracts a disproportionate share of trials due to its broader basin of attraction.

The 9.65-unit energy gap quantifies how much shallower the spurious basins are compared to

the stored attractors. This gap is notably larger than typical Hebbian network demonstrations on symmetric letter sets, indicating that the NOVER pattern set produces a particularly clean energy landscape with well-separated attractors. Spurious states, particularly mixture states formed by overlapping patterns, settle at these intermediate energy levels. Inverted states, mathematically symmetric to the stored patterns, achieve identical global minimum energies.

4.4 The Energy Landscape

To visualize the mechanism of recall, we projected the high-dimensional (100-dimensional) state space of a 2-pattern network into a 2D plane using Principal Component Analysis (PCA).

Figure 6 reveals two distinct populations of configurations. In the PCA scatter (left), the 5000 random states form a dense cloud centered near the origin, with the stored patterns and their inverses located at the periphery. The true depth of the energy wells is more clearly visible in the right panel: the random state distribution peaks just below $E = 0$, while the stored attractors sit at $E = -49.3$, a gap of approximately 49 energy units. This dramatic separation

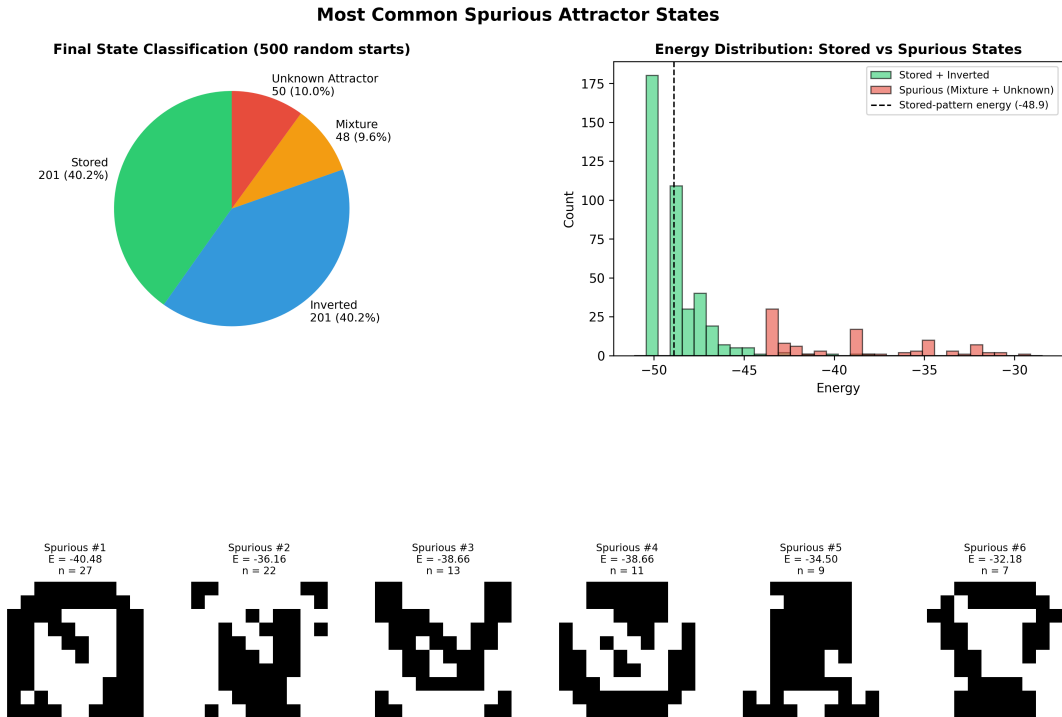


Figure 5: Analysis of 500 recalls from random initial states. (Top-left) Classification of final states. (Top-right) Energy distribution of stored vs. spurious states. (Bottom) Example spurious attractors.

Hopfield Network — Energy Landscape (N=100, P=2: N, O)

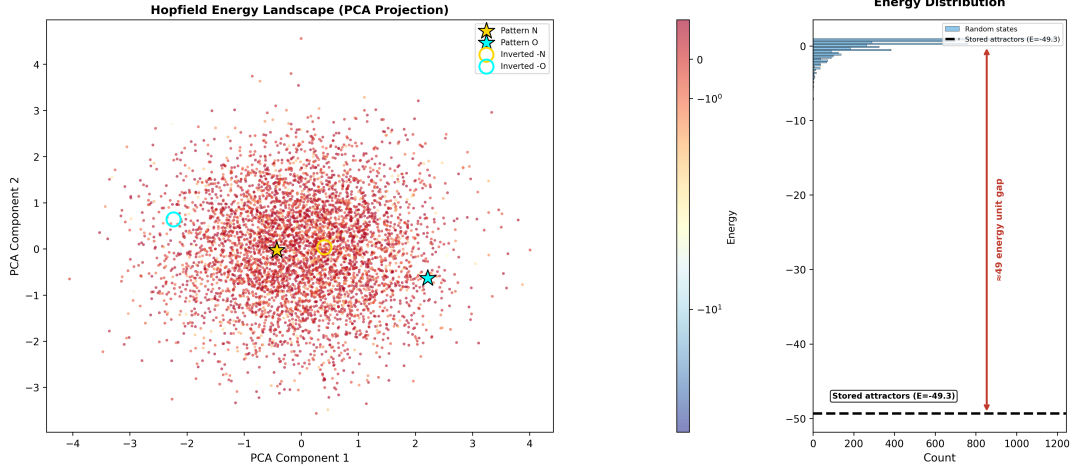


Figure 6: Energy landscape of a 2-pattern network (N, O) projected to 2D via PCA. Left: 5000 random states scatter (colored by symlog energy) with stored patterns (stars) and their inverses (open circles). Right: energy histogram revealing the ~ 49 unit gap between random configurations (clustered near $E \approx 0$) and stored attractors ($E = -49.3$, dashed line).

ration explains why the network is so robust to corruption—moving from a random configuration to a stored pattern releases enormous amounts of energy, and the gradient pulling the system toward the stored state is correspondingly steep.

asynchronous update rule guarantees that every single transition must move to an equal or lower energy contour. This visualizes the Lyapunov property in action: associative memory is literally the physics of a marble rolling downhill.

Hopfield Network — Recall Dynamics (N=100, P=2: N, O)

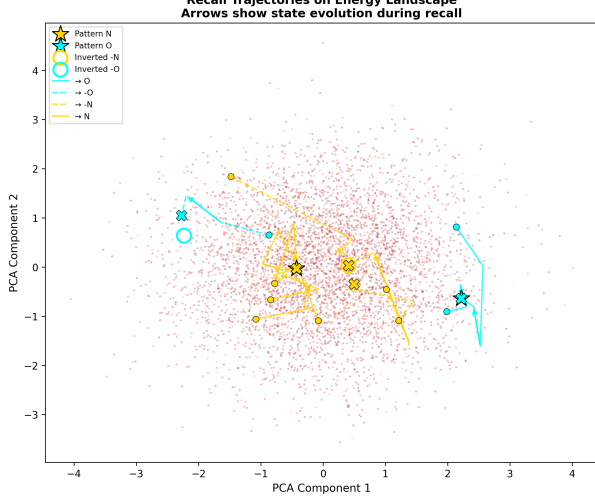


Figure 7: Recall trajectories on the N–O energy landscape. Arrows trace how 10 random initial states evolve during asynchronous recall, converging to one of four attractors: N, O, $-N$, or $-O$. The visit distribution across all four basins is consistent with the expected $\mathbb{Z}_2$ symmetry.

Figure 7 plots actual recall trajectories superimposed on this landscape. The arrows definitively trace gradient descent. The deterministic

4.5 The Ising–Hopfield Isomorphism

The centerpiece of this phase is the formal demonstration that magnetic equilibration and memory recall are identical computational processes. As shown in Figure 8, the trajectory of a Hopfield network correcting a corrupted image implements the same energy-minimization principle as an Ising lattice relaxing toward equilibrium, with the deterministic asynchronous update rule serving as the zero-temperature limit of Metropolis dynamics. Both systems descend a Lyapunov function defined by quadratic spin interactions; the distinction lies only in the source of the coupling matrix (physical exchange constant J versus learned Hebbian weights W_{ij}).

In the Ising model, the system descends an energy gradient defined by a fixed, uniform coupling J , driving the lattice toward a state of uniform magnetization. In the Hopfield network, the system descends a heavily sculpted energy gradient defined by a learned matrix W , driving the network toward a specific, complex configuration.

We summarize this fundamental equivalence in Table 5.

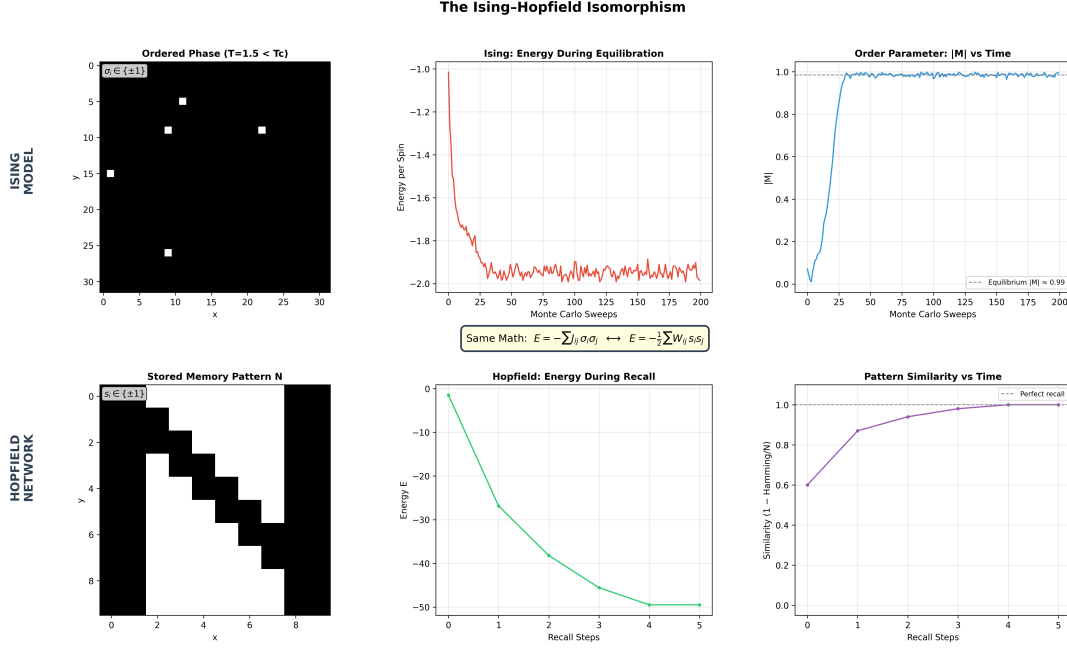


Figure 8: The central result. (Top row) Ising model equilibration: spin lattice, energy descent, order parameter convergence. (Bottom row) Hopfield recall: stored pattern, energy descent, pattern similarity convergence. Both systems minimize the same form of energy function.

Table 5: The Ising–Hopfield isomorphism: a formal mapping between physical and computational quantities.

Ising Model	Hopfield Network
Spin $\sigma_i \in \{\pm 1\}$	Neuron state $s_i \in \{\pm 1\}$
Exchange coupling J	Synaptic weight W_{ij}
External field B	Input bias θ_i
Ground state	Stored memory pattern
Thermal fluctuations	Stochastic updates
Equilibration	Memory recall

The implication is that the Hopfield network does not merely metaphorically resemble physics; it *is* an Ising model with a heterogeneous coupling topology. The ability to retrieve information is an emergent property of statistical mechanics.

5 Discussion

5.1 Quantitative Agreement with Theory

Our experimental results validate the core operating principles of the Hopfield architecture. The network demonstrates robust single-pattern recall, maintaining a $68.0\% \pm 6.7\%$ strict success rate even when a quarter of the input data is destroyed.

However, the capacity analysis (Figure 4) underscores a critical limitation of the standard Hebbian learning rule. While the Amit–Gutfreund–Sompolinsky limit predicts a theoretical capacity of ~ 13.8 patterns for $N = 100$, our network’s accuracy crosses below 50% between $P = 3$ (68.7%) and $P = 5$ (48.8%)—less than half the theoretical limit. This is not a failure of the theory, but rather a consequence of our training data. The theoretical $0.138N$ limit assumes purely orthogonal, random patterns. Natural data—like the letters of the alphabet—are highly correlated. This correlation induces structural interference in the weight matrix, raising the energy floor of the attractors and substantially reducing the effective capacity.

Additionally, the spurious states analysis (Section 4.3) yielded a 201-vs-201 split between stored and inverted convergences, consistent with the $\mathbb{Z}_2$ symmetry that underlies both Hopfield networks and the original Ising model. While this symmetry is mathematically guaranteed by the zero-bias Hebbian energy function, observing near-exact equipartition in a finite sample provides empirical validation of the simulation dynamics.

5.2 Recall as Physical Equilibration

The trajectories mapped in Figure 7 and the direct comparison in Figure 8 confirm our central thesis: associative memory is energy minimization. The network does not “search” for a memory through algorithmic logic; it relaxes into it through thermodynamic necessity. The computational problem of pattern recognition is solved by mapping the desired patterns into the ground states of a simulated physical system.

5.3 Limitations

While theoretically elegant, the classical Hopfield network presented here suffers from several limitations. The $0.138N$ capacity constraint heavily restricts practical utility. The Hebbian learning rule struggles severely with correlated data. Most importantly, the network contains only visible units—it is completely deterministic and operates purely as a retrieval system, entirely incapable of generating new representations or inferring latent features.

6 Conclusion: Bridging to Phase 2.5

Phase 2 has successfully achieved its core objectives, establishing the precise mathematical bridge between the physics of equilibrium thermodynamics and the engineering of associative memory. This paper marks the transition from purely physical simulation to biologically-inspired computation.

6.1 The Limitation: From Recall to Generation

The Hopfield architecture, lacking hidden units, cannot model the latent structure of complex data. It can only recall what it has explicitly memorized. The deterministic update rule, while guaranteeing convergence, traps the system in the nearest local minimum—a desirable property for memory retrieval, but a fundamental obstacle for generative modeling.

6.2 The Solution: Hidden Units and Stochasticity

To transcend memorization and achieve true *generation*, we must introduce two missing elements: unobserved latent variables to cap-

ture complex hierarchical features, and thermal stochasticity to allow the system to escape local minima and sample from a probability distribution [7, 8]. This natural evolution—adding hidden units and temperature to the Hopfield network—yields the Restricted Boltzmann Machine (RBM).

6.3 Summary of Achievements

Phase 2 has successfully accomplished all stated objectives:

- ✓ Implemented Hebbian-trained Hopfield network with asynchronous recall dynamics
- ✓ Quantified single-pattern robustness ($68.0\% \pm 6.7\%$ strict recall at 25% corruption)
- ✓ Mapped storage capacity behavior, observing collapse well below the AGS theoretical limit due to pattern correlations
- ✓ Identified and classified spurious attractors via 500-trial random-initialization analysis
- ✓ Observed empirical $\mathbb{Z}_2$ equipartition (201/201 stored vs. inverted convergences), consistent with theoretical symmetry
- ✓ Visualized the energy landscape via PCA projection, revealing ~ 49 unit energy gap between random states and stored attractors
- ✓ Established the formal Ising–Hopfield isomorphism with side-by-side dynamical comparison

6.4 Future Directions

Phase 2.5 will extend the Hopfield architecture in two directions simultaneously:

- **Hidden units:** introducing latent variables yields the Boltzmann Machine, enabling hierarchical feature representation.
- **Stochasticity:** replacing deterministic sign updates with Glauber dynamics at finite temperature allows the system to escape local minima and sample from a probability distribution.

- **Tractability through restriction:** the Restricted Boltzmann Machine (bipartite visible-hidden graph) makes the resulting model trainable via Contrastive Divergence.
- **Beyond Phase 2.5:** Stacked RBMs lead to Deep Belief Networks, opening potential connections to broader frameworks in statistical field theory that remain to be explored in later phases.

Code and Data Availability

The complete simulation code, analysis scripts, and raw experimental data (including all CSV files referenced in this paper) are available in the project repository at: <https://github.com/YehiaGewily/Physics-AI-Bridge>

Acknowledgments

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